

GLOBAL EHSAN RELIEF · SRI LANKA

GERSL Management System

Complete system report — architecture, modules, workflows, deployment, mobile.

“Serving communities across Sri Lanka.”

Report metadata

Issued	2026-05-04
Audience	NGO leadership, IT operations, prospective deployment partners
Stack	React 19 + Vite · Express + Sequelize · PostgreSQL · Flutter 3.41
Hosting	Vercel (web + serverless API), Supabase Postgres, Cloudflare R2

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1. Executive summary

GERSL is the Global Ehsan Relief – Sri Lanka management system: a single platform that runs every operational facet of the NGO. It combines a React-based web admin for headquarters staff, a Flutter mobile app for field workers, and a shared Node/Express API backed by PostgreSQL on Supabase and S3-compatible object storage on Cloudflare R2. The whole stack is deployed serverlessly on Vercel and is currently in production.

Functionally, GERSL covers the full NGO operating cycle: beneficiary care (orphans, sponsorships, coordinator visits, distributions), programme delivery (projects, activities, tasks, MEAL indicators, evaluations, complaints), field operations (movement register, fuel claims, GPS tracking, live staff map), human resources (staff master, attendance with selfie punch, shifts, onboarding, appraisal, AI-assisted contracts, payroll), finance (chart of accounts, journals, bank/cash accounts, replenishments, bills, payments, fixed assets, donations, grants), procurement (vendor master, RFQ workflow, bid analysis, purchase orders, PDF), fundraising (partners/donors, AI-assisted proposals, campaigns), compliance & safeguarding (incidents, policies, training, background checks, data protection), CBO partner management, public engagement (campaigns, donations, job postings, vendor calls — surfaced on the public portal), and cross-cutting concerns like multi-stage approvals, audit log, notifications, announcements, social media, and reports.

Operationally, the system is one product across three surfaces — web admin, public portal, and mobile — sharing one design language (navy + mission-amber + ink slate, Inter typography) and a deliberate set of primitive components so changes propagate end-to-end. The recent ten-wave UI overhaul standardised every admin page on the canonical navy-band hero pattern and ported the same design system to the Flutter app.

2. What the app does

GERSL replaces what most NGOs piece together from spreadsheets, paper forms, and disconnected SaaS tools. It centralises the day-to-day record-keeping, the approval workflows, the field operations, and the donor-facing public face into one auditable platform.

2.1 Operational scope

- Beneficiary care — orphan registry, sponsorship, coordinator visits, individual & aggregate distributions, needs reports.
- Programme delivery — projects, activities, tasks (assignment + tracking + beneficiary linkage), completion reports.
- Field operations — staff movement register, fuel claims auto-derived from GPS distance, live staff map, shift roster.
- Human resources — full staff master, attendance with selfie + GPS punch + corrections workflow, onboarding checklist, appraisal cycle, AI-drafted contracts, payroll, payslips, leave, salary advances, expense claims.
- Finance — chart of accounts, journal entries, bank accounts/transactions, cash accounts (locker, cash book, petty cash), replenishment queue, bills/payables, payments, invoices, fixed assets, donations, grants.
- Procurement — requisition inbox, vendor master with due-diligence, multi-vendor RFQ + quotation workspace, bid-analysis matrix, purchase order issuance, threshold-based approval routing, PO PDF generation.
- Fundraising — partners/donors, AI-assisted proposal drafting, donor relationship history, campaigns + packages, donation collection.
- MEAL — log-frame indicators, evaluations (wizard-based), complaints, MEAL reports.
- Compliance & safeguarding — incident reporting, policies, mandatory training records, background checks, data-protection requests, learning events.
- CBO partners — CBO master, due diligence, projects, proposals, activities, volunteers.
- Public engagement — public-facing campaign pages, donation collection, job postings, vendor calls (tenders), all editable from the admin and visible on the public portal.
- Cross-cutting — multi-stage Approval Centre, immutable Audit Log, in-app Notifications, organisation-wide Announcements, Social Media planner, Reports & Analytics, Settings (users, roles, permissions, devices).

2.2 Surfaces

Web admin	Used by HQ, finance, HR, programme, procurement, MEAL and management. ~150 pages.
Public portal	12 pages — Home, About, Campaigns (+detail), Careers (+detail), Tenders (+detail), Contact, Orphans-in-Need, Privacy, Terms.
Mobile app	Used by field officers and managers. 14 screens (Today, Visits, Tasks, Movements, Fuel claims, Leave, Expenses, Shifts, Salary advances, Payslips, Approvals, Announcements + Login + Drawer).

3. User personas & roles

Every action in GERSL is permission-gated. Roles are defined in the database and each one is mapped to a flat key-space of 150+ permissions of the form ``module:action`` (e.g. ``orphans:view``, ``finance:approve_po``, ``projects:ceo_approve``). A single super-admin permission ``*`` exists for emergency access. The wildcard fallback bypasses page-level checks entirely.

3.1 Built-in roles

Role	Typical user	Primary responsibilities
Admin / Super-admin	IT	Full system access, manage users, roles and permissions.
BOD	Board members	Read-only oversight across modules; sees Executive Dashboard.
CEO	Executive Director	Final approver on high-value POs, projects and proposals.
Director Programmes	Senior management	Approves project plans, oversees programme delivery.
Programme Manager	Programme leads	Manages projects, tasks, MEAL indicators and field teams.
Project Officer (multiple variants)	Field/admin staff	Executes activities, logs visits, distributes aid.
Finance Manager	Finance head	Approves bills, journals and payroll runs.
Finance Officer / Accountant	Finance team	Records transactions, reconciles cash and bank.
HR Manager	HR head	Approves leave, manages onboarding, payroll, contracts.
HR Officer / Assistant	HR support	Records attendance corrections, manages shifts.
Fundraising Manager	Fundraising head	Manages donors, proposals and campaigns.
MEAL Officer	MEAL team	Manages indicators, evaluations and complaints.
Media / Media Production Officer	Comms team	Runs the social media planner and announcements.
Field Officer	Field staff	Mobile-first: punch, visits, tasks, claims, expenses.
Orphan Coordinator	Field staff	Manages assigned orphan caseload and visit scheduling.
Guest	External	Read-only access to specific shared dashboards.

3.2 How permissions cascade

On ``/auth/login``, the API returns the user record together with their flattened permission list. The web app stores the user in ``AuthContext`` and all pages, sidebar items and action buttons gate themselves with ``hasPermission('module:action')``. The mobile app keeps the same user object in ``flutter_secure_storage`` and gates approval/admin features the same way (see ``_approverRoles`` whitelist for the mobile Approvals queue). Server routes additionally re-check permissions in their middleware so a manipulated client cannot escalate.

4. Module-by-module walkthrough

• Dashboard

Three dashboards: Executive (KPIs across the org), Admin (operational health), Staff "My Dashboard" (personal tasks, visits, leave balance). Lives Reached and mission-amber metrics are surfaced in hero cards.

• Orphan Care

Master list with grid/list/map views, sponsorship records, coordinator assignment, visit logs, individual & aggregate distribution flows, needs report wizard, bulk Excel upload.

• Coordinators

Field coordinator roster, assignments, performance, communication log.

• Beneficiaries

Universal beneficiary database shared across projects. Bulk upload, list generator (Annex-style), demographic filters, support history.

• Partners & Donors

Partner/donor master, communication log, contribution analytics, contract documents.

• Proposals

Multi-stage proposal pipeline (Draft !' Internal Review !' CEO !' Donor !' Funded). AI assistant drafts narrative sections from a brief. Email modal pushes to donor.

• Projects

Portfolio view with budget distribution, status overview, completion-rate trend, performance KPIs. Per-project: team members, financials, tasks (Gantt + list), MEAL indicators/evaluations/complaints, completion report (Annex D).

• Activities

Within projects: scheduled activities with deliverables, beneficiary counts and media coverage uploads.

• Tasks / My Tasks

Org-wide task board with assignment, comments, attachments and beneficiary linkage. Field staff see only "My Tasks".

• Movement register

Field-trip log: plan !' depart !' arrive !' return. Auto-tracks distance for fuel-claim derivation.

• Fuel claims & rates

Auto-derived from returned movements (rate × km, lunch-hour deduction). Vehicle-typed rates effective-dated.

• Approvals

Centralised queue across proposals, projects, finance items, leave, expenses, salary advances. Mobile mirror for managers on the go.

• Compliance & Safeguarding

Incidents, policies, training records, background checks, data-protection requests, learning events.

• Finance

Chart of accounts, journal entries, bank + cash accounts, locker / cash book / petty cash registers, replenishment queue, bills/payables, payments, invoices, fixed assets, financial reports.

• Procurement

Inbox of requisitions assigned to procurement officers, vendor master with due-diligence, RFQ workspace (invite vendors, capture quotations, bid-analysis matrix), purchase order issuance with PDF, scope-aware threshold matrix.

- **Public Engagement**

Campaigns (with packages) !' donations !' public donor flow. Job Postings !' public Careers page. Vendor Calls !' public Tenders page.

- **CBO Partners**

CBO master, due diligence, projects, proposals, activities, volunteers — separate workflow from internal projects.

- **MEAL**

Indicators with progress tracking, evaluation wizards, complaint intake.

- **Social Media**

Content calendar, scheduled posts, analytics cards.

- **Announcements**

Org-wide messaging visible in admin and on the mobile drawer feed.

- **Reports & Analytics**

Cross-module reports with PDF/Excel export and email scheduling.

- **Settings**

User management, role/permission editor, device management, notification rules, system settings.

5. End-to-end workflows

5.1 Project !' activity !' task !' distribution

- Programme Manager creates a Project with budget, timeline, programme area, location.
- Director Programmes / CEO approve in the Approvals Centre (multi-stage gate).
- Project Officer creates Activities under the project (with planned dates, deliverables, target beneficiaries).
- Tasks are assigned to staff with priorities, due dates and beneficiary linkage.
- Field staff execute and record either an Individual Distribution (per beneficiary) or an Aggregate Distribution (head-count + media coverage).
- MEAL indicators auto-update from underlying records; evaluations and complaints attach to the project.
- On closure, Project Completion Report (Annex D) is generated as PDF.

5.2 Procurement (RFQ !' PO)

- Requisition created (often from a project or budget line) lands in Procurement Inbox.
- Procurement Manager assigns to an officer and selects the procurement method.
- Officer opens RFQ Builder Modal, picks "e3 active vendors from the master, sets scope-of-work, closing date, payment & delivery terms.
- Draft RFQ created !' Send marks it Sent and emails each invited vendor.
- Quotations captured in the RFQ Workspace; bid-analysis matrix computes recommended vendor.
- PO drafted !' routed through threshold matrix (scope: global / donor / project, by amount band) for approval.
- On approval, PO PDF generated server-side via pdfkit and sent to vendor.

5.3 Field punch !' fuel claim

- Field officer opens the mobile app, hits "Punch IN" on the Today screen.
- App captures front-camera selfie + GPS coordinates + accuracy !' uploaded to R2 !' recorded with geofence-match flag.
- Officer logs a Movement (plan !' depart !' arrive !' return) over the day.
- On Return, distance × vehicle-type rate (minus lunch-hour deduction) auto-creates a Fuel Claim.
- Approvers see pending claims under Operations · Fuel Claims and the mobile Approvals queue.

5.4 Leave / expense / advance request

- Staff submits via mobile (or web) — request lands in Pending status.
- Notification fires to the appropriate approver pool (HR Manager for leave, Finance Manager for expenses & advances).
- Approver opens Approvals Centre (web or mobile), reviews details, approves or rejects with comment.
- Approved expenses flow into the next payable run; approved advances are deducted from the next payslip.

5.5 Donor proposal

- Fundraising Officer creates a draft proposal — fills donor, project tie-in, requested amount, narrative outline.
- AI Assistant drafts narrative sections from a brief.
- Internal review !' CEO sign-off !' Donor submission via the integrated email modal.
- Pipeline view tracks each proposal's stage and success rate.
- On Funded status, a Project shell can be auto-created and budget seeded from the proposal.

6. Mobile app — field workflow

The Flutter mobile app is built for field staff and managers, designed to function on low-end Android devices and intermittent connectivity. It uses the same backend, the same auth, and (after the recent overhaul) the same design language as the web admin.

6.1 Daily field workflow

- Open app !' already signed in via secure storage; landing tab is "Today".
- Today screen shows a navy hero card with current clocked-in status, last punch time, and primary Punch IN (mission-amber) / Punch OUT (outline) buttons.
- On Punch IN, the app captures a selfie + GPS, uploads via the punch endpoint, and records the geofence-match.
- Throughout the day: log Visits (beneficiaries served, project, purpose), tick off Tasks, register Movements (depart/arrive/return), submit Expenses and Leave requests.
- Background GPS service with foreground notification keeps tracking on for assigned routes; toggle from the app bar.
- Managers also see the Approvals queue (leave, expense, advance) for their team, and the Announcements feed.

6.2 Mobile screens

Surface	Purpose
Today / Punch	In/Out + Break + selfie + GPS, today's punch list.
Visits	List + log visits with beneficiary count, project, purpose.
My Tasks	Assigned tasks with priority + status + project linkage.
Movements	Field-trip register: plan !' depart !' arrive !' return.
Fuel claims	Auto-derived claims from returned movements.
Leave	Request + cancel pending; statuses visible.
Expenses	Submit with receipt photo; cancel pending.
Shifts	Roster for the next 60 days.
Salary advances	Request; managers can approve/reject in-line.
Payslips	List + open PDF externally.
Approvals	Manager-only batched queue (leave + expense + advance).
Announcements	Read-only org-wide feed.

6.3 Mobile-specific permissions

The mobile app declares: INTERNET, ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, ACCESS_BACKGROUND_LOCATION, CAMERA, FOREGROUND_SERVICE, FOREGROUND_SERVICE_LOCATION, WAKE_LOCK, RECEIVE_BOOT_COMPLETED, POST_NOTIFICATIONS. The background location service uses foregroundServiceType="location" so Android 12+ permits continuous tracking with a persistent notification.

7. Architecture overview

GERSL is a single-repo monorepo deployed as one Vercel project. The frontend (web admin + public portal) is a single-page React app. The backend is the same Express app you would `node` locally — but invoked from Vercel's serverless runtime via `api/index.js` which directly calls `app(req, res)` (the `serverless-http` wrapper was removed because it caused 30-second hangs).

7.1 Request paths

- Browser hits Vercel edge !' `/api/\*` rewritten to the serverless function `api/index.js` !' Express app !' controller !' Sequelize !' Postgres on Supabase.
- Browser hits Vercel edge !' static path !' CDN serves Vite-built JS/CSS/HTML from `dist`.
- Browser hits any unknown path !' SPA fallback rewrite serves `index.html`, React Router takes over.
- Mobile app (dio) !' Same `/api/\*` endpoints. Auth via Bearer header from secure storage.
- File uploads !' multer-s3 !' Cloudflare R2 (S3-compatible). URLs returned to client.
- Daily cron !' Vercel cron triggers `api/cron/cluster` at 01:30 UTC for movement clustering.

7.2 Logical layers

Presentation	React 19 + Tailwind 3 + Radix UI; Flutter 3.41 + Material 3 + Riverpod.
State	Web: Zustand + React Contexts + TanStack Query. Mobile: Riverpod.
API contract	REST under `/api`; JWT in httpOnly cookie (web) and Bearer header (mobile).
Domain	Express controllers + Sequelize models; multi-stage Approval domain.
Persistence	Postgres on Supabase (managed, SSL, IPv4); SQLite fallback for offline dev.
Object store	Cloudflare R2 via @aws-sdk/client-s3 + multer-s3 + signed URLs.
Background	node-cron + Vercel scheduled cron; foreground GPS service in Flutter.
Observability	morgan request logs + winston app logs; immutable AuditLog model.

8. Tech stack — web

Concern	Choice
Framework	React 19.1 + Vite 7.1
Routing	react-router-dom 7.9
Styling	Tailwind 3.4 with custom design tokens (navy, ink, mission, success, danger)
Typography	Inter + Inter Tight via Google Fonts
Component primitives	src/components/ui/primitives.jsx — PageHeader, PageWrap, Card, Button, Badge, StatusBadge, EmptyState, ErrorBox, Field, Th/Td
Headless UI	Radix (Dialog, Dropdown, Select, Tabs, Tooltip, Popover, Switch, Checkbox, Radio)
Icons	lucide-react
State	Zustand 5 + 22 React Contexts + TanStack React-Query 5.90 + react-error-boundary
HTTP	axios; base URL via src/config/apiBase.js
Validation	zod
Maps	leaflet + react-leaflet
Charts	recharts
Tables / exports	xlsx, file-saver, jsPDF, jsPDF-autotable
Sanitization	dompurify
Toasts	react-toastify
Build chunking	Manual: react-vendor, ui-vendor, utils-vendor, chart-vendor, document-vendor, plus per-module chunks
Minification	terser, drop_console + drop_debugger in prod
ESLint	flat config + react-hooks + react-refresh
Code volume	~103,661 LOC across 60 page files + 51 page-level components + 22 contexts + ~200 shared components

9. Tech stack — backend

Concern	Choice
Runtime	Node.js (ESM "type":"module")
Web framework	Express 4.18
ORM	Sequelize 6.35 with pg-hstore
Database	PostgreSQL on Supabase (project gersl-management-dev, ap-northeast-1). SQLite fallback for offline dev.
Static pg import	dialectModule: pg so Vercel NFT can trace it (Sequelize's dynamic require fails on serverless).
SSL	Auto-on in prod or DB_SSL=true; family:4 forced to dodge IPv6.
Auth	JWT (jsonwebtoken) — httpOnly cookies for web, Bearer header for mobile.
Hashing	bcryptjs
CSRF	csrf-csrf
Validation	express-validator + zod
Security	helmet, express-rate-limit, cors, cookie-parser
Logging	morgan (HTTP) + winston (app)
File uploads	multer (local) + multer-s3 + @aws-sdk/client-s3 !' Cloudflare R2
Pre-signed URLs	@aws-sdk/s3-request-presigner
PDF	pdfkit (PO docs, payslips, completion reports, this report)
Email	nodemailer
Cron	node-cron + Vercel scheduled cron
AI	pluggable services in server/src/services/ (Puter / Gemini variants)
Code volume	~51,067 LOC across 84 routes, 106 controllers, 20 root models (+ many domain models), 7 services, 8 middleware

10. Tech stack — mobile

Concern	Choice
SDK	Flutter 3.41.2 stable (engine d96704abcce, 2026-02-18)
Dart SDK	^3.11.0
App ID	org.gersl.gersl_mobile
Version	1.0.0+1
Min SDK	Android 21 (5.0) — required by background service & local notifications
Target SDK	Inherited from Flutter (35 / Android 15)
Java target	17 with core-library desugaring (desugar_jdk_libs:2.1.4)
State / DI	Riverpod 2.5
Routing	go_router 14.6
HTTP	dio 5.7 with auth interceptor
Persistence	flutter_secure_storage (tokens) + shared_preferences (settings)
Permissions	permission_handler 11.3
Geo / Background GPS	geolocator 13 + flutter_background_service 5.0 (foregroundServiceType=location)
Camera	image_picker 1.1 (front-camera selfies on punch-in)
Notifications	flutter_local_notifications 17.2
URLs	url_launcher 6.3
Formatting	intl 0.20
Typography	google_fonts 6.2 (Inter, matches admin web)
Design tokens	mobile/lib/app/theme.dart — kNavy900 #0D1D3D, kMission500 #F59E0B, kInk* slate scale
Primitives	mobile/lib/app/widgets.dart — SoftCard, GlassCard, GlowButton, StatusPill, GradientBlob, GradientText, SectionHeader, MobilePageHeader, EmptyState, ErrorBox, LoadingPanel
Splash + adaptive icon	flutter_native_splash + flutter_launcher_icons; colour #0D1D3D
Code volume	~4,998 LOC across 36 Dart files
Release APK	54.1 MB, debug-signed (sideload only)

11. Data model

The schema is large (~70 tables). Here is the core model graph; many other domain models live alongside their feature controllers (cash accounts, journal entries, RFQs, POs, vendor calls, beneficiary support, fuel claims, leave balances, payslips, location points, movement clusters, CBO entities, etc.).

11.1 Core entities

- User — the staff record + auth identity. Belongs to Department, has Role + Permissions.
- Department, Position — org structure.
- Project — programme delivery unit. Has Budget, Team Members, Tasks, Activities, Indicators, Evaluations, Complaints, Distributions.
- ProjectTeamMember — User × Project assignment.
- Task — Work unit under a Project. Has Assignees, Comments, Attachments, Beneficiaries.
- Beneficiary — universal record across projects (orphan, family, individual).
- Orphan — specialised Beneficiary with sponsorship + visit log.
- AggregateDistribution — head-count distributions with media coverage.
- Approval — Polymorphic approval record across proposals, projects, finance items, leaves, expenses, advances.
- Notification — in-app + push messaging.
- AuditLog — append-only record of significant state changes.
- Complaint, Indicator, Report — MEAL primitives.
- StaffDocument — file attachments on staff records.
- Expense (Mobile-first) — staff out-of-pocket claims.
- Proposal — fundraising pipeline record with stages.

11.2 Storage

Primary DB	PostgreSQL on Supabase (jkmkeycwpqvhgjmkyap.supabase.co), region ap-northeast-1.
Object store	Cloudflare R2 — staff documents, orphan photos, PO PDFs, selfies, receipts, completion reports.
Backup posture	Supabase managed backups (daily). PITR depends on plan tier.
Migrations	Sequelize sync + manual scripts in server/src/database/.

12. Security & RBAC

12.1 Auth flow

- POST /api/auth/login with username/password.
- Server validates with bcrypt; on success issues JWT (1h) + refresh token, sets as httpOnly Secure cookies (web) and returns in body for mobile to store in flutter_secure_storage.
- All subsequent requests carry the JWT (cookie or Bearer header). middleware decodes, attaches `req.user` (id, role, permissions list).
- Logout clears the cookies + secure storage and revokes the refresh token server-side.

12.2 Defence in depth

- Helmet sets a tight Content-Security-Policy by default.
- express-rate-limit on /auth/login + sensitive write endpoints to slow brute force.
- csrf-csrf double-submit pattern for browser-originated mutations.
- CORS limited to the deployed origin in prod.
- All input validated with express-validator / zod at the route boundary.
- Sequelize parameterised queries everywhere — no raw SQL concatenation.
- Permissions re-checked server-side on every state-changing endpoint.
- Audit Log appends user id, action, target id and diff for every significant change.
- Selfies + receipts uploaded via signed pre-uploaded URLs — no direct R2 credentials in the client.
- JWT signing key rotated per-environment via JWT_SECRET env var.

12.3 RBAC keyspace (excerpt)

Module	Sample permissions
Dashboard	dashboard:view
Orphans	orphans:view, orphans:create, orphans:edit, orphans:delete, orphans:approve, orphans:assign_coordinator, orphans:sponsorship_manage
Projects	projects:view, projects:create, projects:edit, projects:approve, projects:director_approve, projects:ceo_approve
Finance	finance:view, finance:create_expense, finance:approve_po, finance:manager_approve, finance:ceo_approve, finance:reconcile_accounts, finance:cash:*
HR	hr:view, hr:approve_leave, hr:manager_approve_leave, hr:attendance_manage, hr:onboarding, hr:appraisal, hr:manage_contracts, finance:view_payroll
Procurement	procurement:dashboard_view, procurement:request_view, procurement:vendor_view, procurement:thresholds_manage
Wildcard	* (super-admin)

13. Design system

GERSL has one design language across web admin, public portal, and mobile. The recent ten-wave overhaul standardised every surface on the same tokens and primitive components.

13.1 Tokens

Token	Web (Tailwind)	Mobile (Dart)	Hex
Primary	navy-900	kNavy900	#0D1D3D
Primary darker	navy-800	kNavy800	#142A55
Accent	mission-500	kMission500	#F59E0B
Accent on dark	mission-300	kMission300	#FCD34D
Text primary	ink-900	kInk900	#0F172A
Text secondary	ink-500	kInk500	#64748B
Border	ink-200	kInk200	#E2E8F0
Surface canvas	ink-50	kInk50	#F5F7FA
Card	white	Colors.white	#FFFFFF
Success	success-600	kSuccess600	#16A34A
Danger	danger-600	kDanger600	#DC2626

13.2 Canonical page anatomy

- Web: `<PageWrap>` (p-6 max-w-7xl mx-auto space-y-4) !' `<PageHeader` icon eyebrow title subtitle actions' (navy band) !' `<Card>`'s.
- Mobile: `Scaffold` !' `MobilePageHeader` (matching navy band) !' `SoftCard` rows with `StatusPill` + `EmptyState` / `ErrorBox` / `LoadingPanel` for async states.
- Action button hierarchy: primary (navy) for permanent actions; mission-amber for hero CTAs (Add, New, Sign in); onNavy outline for secondary actions inside the navy band.

13.3 Typography

Inter across all surfaces, from Google Fonts (web CDN, mobile via google_fonts package). Headings use negative letter-spacing and weight 800 for the tight, considered SaaS feel; body uses weight 400/500 with 1.4 line-height for readability.

14. Deployment & operations

14.1 Topology

Layer	Provider	Notes
Web frontend	Vercel CDN	Static dist/ from `vite build`. Manual chunking by route group.
API	Vercel serverless	Single function api/index.js ! Express app (30 s maxDuration).
Database	Supabase Postgres	Project gersl-management-dev, region ap-northeast-1, SSL on, IPv4 forced.
Object storage	Cloudflare R2	S3-compatible; orphan photos, staff docs, PO PDFs, selfies, receipts.
Cron	Vercel cron	/api/cron/cluster daily at 01:30 UTC for movement clustering.
Source control	GitHub	jarooq/gersl-management. Push to main triggers Vercel build.
Mobile distribution	Sideload APK	Release-signed keystore TODO before Play Store.

14.2 Environment variables

- DATABASE_URL — Supabase Postgres connection string.
- JWT_SECRET — JWT signing key.
- R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_BUCKET, R2_ENDPOINT — Cloudflare R2.
- EMAIL_HOST, EMAIL_PORT, EMAIL_USER, EMAIL_PASS — SMTP for nodemailer.
- AI_PROVIDER, AI_API_KEY — AI provider routing.
- VERCEL=1 — set automatically; gates serverless-aware code paths.

14.3 Production fixes already shipped

- Static `import pg from "pg"` in database.js — Vercel's NFT could not trace Sequelize's dynamic require.
- Wrapped `mkdirSync` calls in try/catch — Vercel filesystem is read-only.
- Replaced serverless-http with direct `app(req, res)` — wrapper caused 30 s 504 hangs.
- SPA catch-all rewrite added — `/(?!assets/|.\*/[a-zA-Z0-9]+\$).\*/` ! ` /index.html`.
- Hoisted multer-s3, @aws-sdk/*, node-cron, pdfkit from server/package.json to root — Vercel installs only the root manifest.
- Escaped backslash in vercel.json SPA rewrite regex (`\\.`) — invalid JSON had been rejecting every deploy for an hour.
- Cash-module permissions re-seeded — frontend referenced finance:cash:* but seed had not created them.

15. Project status, KPIs & metrics

15.1 Codebase scale

Surface	Files	Lines of code
Web admin + public portal	60 page files + 51 page-components + 22 contexts + ~200 shared components	103,661
Backend API	84 routes + 106 controllers + 20 root models + 7 services + 8 middleware	51,067
Mobile (Flutter)	36 Dart files (3 in app/, 11 features, 4 services)	4,998

15.2 Production health

- Latest deploy: gersl-management-omq9bewj5 — Ready (45 s build).
- Previous 10 deploys errored due to invalid escape in vercel.json — fixed.
- flutter analyze on mobile: No issues found.
- npm run build on web: 20 s, no errors. Bundle warning on document-vendor (796 KB jspdf+xlsx).

15.3 Recent work (last 20 commits)

- Mobile redesign — full theme port + 11 screens reskinned (5e4cde3).
- UI waves 1–10 (~150 admin pages migrated) — design tokens, primitives, sidebar/navbar/login/layout, dashboards, HR, Finance, Operations, Procurement, Beneficiaries, Coordinators, Orphans, Partners, CBO, Proposals, Approvals, MEAL, Social, Donations, Campaigns, Jobs, VendorCalls, Projects, Reports, Settings.
- Approvals + Contracts + Payroll + MEAL + Social Media polish (eb37853).
- vercel.json fix — escaped backslash (ae4d8a3).
- Cash module permissions seeded (7828890).
- Layout fix — brand block in sidebar, removed duplicate from navbar (d6e8dc3).

16. Roadmap & known follow-ups

Blocker

GERHR (legacy Firebase HR app) data migration into GERSL is paused awaiting the Firebase service-account JSON for project gerhr-388e1. Once supplied, the migration script can be wired and the GERHR app sunset.

16.1 Short-term

- Generate a release keystore for the mobile app (`signingConfigs.create("release")` in `build.gradle.kts`) before Play Store upload.
- Decide whether `usesCleartextTraffic="true"` should be removed from production builds (currently kept for the dev/staging API).
- Code-split the document-vendor chunk (jspdf + xlsx) using dynamic imports to drop the 796 KB bundle warning.
- Wire push notifications via `firebase-admin` (server-side already imports it).
- Bump the mobile `versionCode` on every Play Store release.

16.2 Medium-term

- Replace `Sequelize sync` with proper migrations (`Umzug` or `sequelize-cli`).
- Move file uploads to direct-to-R2 pre-signed URLs to bypass the Vercel function size limits.
- Add e2e tests (Playwright) for the critical workflows (login, RFQ !' PO, approve leave).
- Surface a public-facing donation receipt PDF flow.
- Add SMS notifications for field staff (Sri Lankan SMS gateway integration).

16.3 Long-term

- Multi-tenancy if the platform expands to other Ehsan Relief country offices.
- Offline-first capability on mobile (drift / `sqlite` + sync queue) for low-connectivity field work.
- Public donor portal — donors log in to see what their money funded.
- External donor reporting API (push project completion + spend to donor systems).

17. Appendix — admin pages, routes, models

17.1 Admin sidebar (43 items)

Admin page	Route
Dashboard	/admin/dashboard
My Dashboard	/admin/my-dashboard
Orphan Management	/admin/orphans
Coordinators	/admin/coordinators
Beneficiaries	/admin/beneficiaries
Partners	/admin/partners
Proposals	/admin/proposals
Projects	/admin/projects
Activities	/admin/operations/activities
All Tasks	/admin/operations/tasks
My Tasks	/admin/operations/my-tasks
Movement Register	/admin/operations/movements
Fuel Claims	/admin/operations/fuel-claims
Fuel Rates	/admin/operations/fuel-rates
Approvals	/admin/approvals
Compliance & Safeguarding	/admin/compliance
Finance	/admin/finance
Cash Accounts	/admin/finance/cash/accounts
Cash Replenishments	/admin/finance/cash/replenishments
Campaigns	/admin/campaigns
Donations	/admin/donations
Job Postings	/admin/job-postings
Vendor Calls	/admin/vendor-calls
Procurement Dashboard	/admin/procurement/dashboard
Procurement Inbox	/admin/procurement/inbox
Procurement Vendors	/admin/procurement/vendors
Procurement Thresholds	/admin/procurement/thresholds
HR Overview	/admin/hr
Attendance	/admin/hr/attendance
Attendance Corrections	/admin/hr/corrections

Admin page	Route
Live staff map	/admin/hr/live-map
Shift roster	/admin/hr/shifts
Movement segments	/admin/hr/movement-segments
Onboarding	/admin/hr/onboarding
Appraisal	/admin/hr/appraisal
Contracts	/admin/hr/contracts
Payroll	/admin/hr/payroll
CBO Partners	/admin/cbo
MEAL	/admin/meal
Social Media	/admin/social-media
Announcements	/admin/announcements
Reports	/admin/reports
Settings	/admin/settings

17. Appendix (cont.)

17.2 Backend route surfaces (84)

aggregateDistribution	ai	announcement	approval
attendance	auditLog	auth	background-check
bankAccount	bankTransaction	beneficiaries	beneficiarySupport
bill	budget	budgetCategory	campaign
campaignPackage	cash	cbo	cbo-activity
cbo-due-diligence	cbo-project	cbo-proposal	cbo-volunteer
chartOfAccounts	complaint	compliance	compliance-training
contract	coordinator	data-protection	department
device	donation	donor	evaluation
finance	financialReport	fixedAsset	grantReceipt
grantReceivable	hr	hr-appraisal	hr-onboarding
invoice	jobPosting	journalEntry	learning-event
leaveBalance	locationPoint	meExpense	meLeave
mePayslip	meal	movement	movementCluster
notification	orphan	partner	payable
payment	payroll	position	procurement
project	projectTeam	proposal	purchaseOrder
report	role	safeguarding-incident	safeguarding-policy
salaryAdvance	shift	socialMedia	staffDocument
task	taskBeneficiary	upload	uploads-static
users	vendorCall	visit	visitLog

17.3 Root models (20)

AggregateDistribution	Approval	AuditLog
Complaint	Department	Expense
Indicator	Notification	Project
ProjectTeamMember	Proposal	Report
StaffDocument	Task	TaskAssignee

TaskAttachment	TaskBeneficiary	TaskComment
User		

Many additional domain models live in subfolders alongside their controllers (cash accounts, journal entries, RFQs, POs, vendor calls, beneficiary support, fuel rates / claims, leave balances, payslips, location points, movement clusters, CBO entities, etc.) — the comprehensive index lives in `server/src/models/index.js`.